

CivicPulse Monsoon

AI-Based Predictive Waterlogging & Drainage Risk Management System

Target Domain: Municipal Waterlogging & Urban Disaster Support

Event: Smart India Hackathon (SIH)

Status: Production-Ready Prototype

1. Executive Overview & Core Objective

CivicPulse Monsoon is an end-to-end AI-powered geospatial platform engineered for municipal corporations and urban disaster management authorities. Existing municipal systems operate *reactively* — citizen complaint apps record waterlogging only after streets flood, paralyzing traffic and impeding emergency services.

CivicPulse Monsoon shifts municipal operations from reactive firefighting to **proactive predictive management**.

Feature / Aspect	Traditional Municipal Systems	CivicPulse Monsoon Platform
Operational Mode	Reactive (Post-event complaint logs)	Proactive (24h/48h/72h predictive forecasting)
Risk Insights	Historical complaints only	SHAP Explainable AI rationale for risk drivers
Spatial Awareness	Isolated ward points	Drainage Dependency Graph & PostGIS Overlaps
Resource Allocation	Manual / ad-hoc crew dispatch	Google OR-Tools MILP Crew & Budget Optimizer
Scenario Planning	None (Trial and error during storms)	Interactive What-If Scenario Simulator

2. System Architecture & Tech Stack

The architecture is decoupled into a high-performance **FastAPI Backend**, a geospatial **PostgreSQL/PostGIS Database**, a machine learning & optimization engine, and a **Next.js 14 Web Command Center**.

Layer	Technologies	Key Purpose & Responsibilities
Frontend App	Next.js 14, React 18, TypeScript, Tailwind CSS, Leaflet	Interactive Command Center map, SHAP visual charts, resource planners, and simulator dashboard.
Backend REST API	Python 3.12, FastAPI, Uvicorn, Pydantic v2	High-throughput REST API serving predictions, spatial GeoJSON layers, priorities, and optimizations.
Database & GIS	PostgreSQL 15 + PostGIS 3.3, SQLAlchemy 2.0, GeoAlchemy2	Spatial data store for wards, roads, drainage channels, waterbodies, and historical incidents (with SQLite fallback).
ML & Explainability	XGBoost, LightGBM, scikit-learn, SHAP, Joblib	Multi-horizon risk probability classification (24h/48h/72h) and SHAP directional feature attribution.
Optimization Solver	Google OR-Tools (MILP Solver)	Optimal municipal crew and budget allocation under capacity, skill, and location constraints.

3. Core Platform Modules (10 Functional Systems)

1. Geospatial Command Center Map

Interactive map interface supporting 8 switchable GIS layers: Wards, Roads, Drains, Waterbodies, Historical Incidents, Critical Facilities, Population Heatmaps, and Risk Propagation dependencies.

2. Historical Pattern Analysis Engine

Calculates rolling 1h to 72h rainfall aggregations, 7d/30d/90d incident counters, and ward-level recurrence hotspot metrics (HIGH / MEDIUM / LOW).

3. Rainfall Analytics & Gauge Tracker

Monitors rainfall intensity, spatial interpolation across municipal zones, and forecasts versus observed rainfall deviations.

4. Multi-Horizon Risk Prediction Models

Versioned XGBoost & LightGBM classifiers predicting waterlogging event probabilities for 24-hour, 48-hour, and 72-hour forecasting windows.

5. SHAP Explainable AI (XAI)

Generates directional feature contribution progress bars and human-readable evidence summaries explaining exact rationale for high risk predictions.

6. Drainage Network Dependency Graph

Spatial graph modeling topological relationships between drainage channels, road culverts, low-lying zones, and critical infrastructure to locate bottlenecks.

7. Population & Infrastructure Exposure Estimator

PostGIS spatial overlap calculations estimating exposed population density and counts of nearby critical facilities (hospitals, schools, transit).

8. Civic Priority Engine

Configurable weighted scoring matrix assigning municipal urgency levels: P1 — CRITICAL, P2 — HIGH, P3 — MEDIUM, and P4 — LOW.

9. Municipal Crew & Budget Optimizer

Google OR-Tools MILP solver allocating maintenance crews, working hours, equipment, and budgets to maximize risk reduction under strict operational bounds.

10. What-If Scenario Simulator

Multi-stage analytical simulation engine evaluating hypothetical storm intensity spikes and resource re-allocations without altering live database records.

4. Project Structure & API Endpoints

The project repository is structured into distinct, modular subsystems:

- **backend/**: FastAPI application, database schemas, ORM models, and service business logic.
- **frontend/**: Next.js 14 application with MapLibre/Leaflet rendering and analytics dashboards.
- **ml/**: Training pipelines, dataset generators, model artifacts, and SHAP explainers.
- **pipeline/**: ETL and data processing scripts.
- **tests/**: Comprehensive automated test suite with over 44 unit and integration tests.

Endpoint Group	REST Path	Description
Health	/health	System status & DB connectivity check
Predictions	/api/v1/predictions	24h, 48h, 72h risk probability predictions
XAI Explanations	/api/v1/explanations/{ward_id}	SHAP feature importance & rationale
Spatial GeoJSON	/api/v1/geojson/{layer}	GIS vector layers for map UI
Dependency Graph	/api/v1/graph	Drainage network graph nodes & edges
Priority Scoring	/api/v1/priority	P1-P4 municipal urgency ranking
Optimization	/api/v1/optimization	OR-Tools crew & budget allocation plan
Simulation	/api/v1/simulation	What-If scenario simulation execution

5. Local Execution & Automated Verification

Backend Startup (Powershell):

```
cd backend
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
$env:PYTHONPATH="..;"
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

Frontend Startup:

```
cd frontend
npm install
npm run dev
```

Automated Test Suite (44+ Pytest Tests):

```
$env:PYTHONPATH="..;backend"; .\backend\venv\Scripts\pytest.exe tests/
```

6. Data Provenance & Decision Support Disclaimer

- **Data Provenance:** Spatial boundaries (wards, roads, elevation) form the core geographic baseline. Synthetic demonstration records carry explicit tags: *'DEMO PROTOTYPE — SYNTHETIC DATA & HISTORICAL CIVIC DATA'*.
- **Decision Support Scope:** Machine learning predictions, priority rankings, and OR-Tools optimization outputs serve as analytical decision-support signals for municipal engineers. They assist in resource prioritization and do not replace physical site inspections.
- **Simulation Isolation:** All What-If scenario simulations execute in an isolated analytical memory layer and never overwrite or mutate live production database records.

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